Topic 1 – Key concepts in chemistry

Atomic structure

Students should:		Maths skills
1.1	Describe how the Dalton model of an atom has changed over time because of the discovery of subatomic particles	
1.2	Describe the structure of an atom as a nucleus containing protons and neutrons, surrounded by electrons in shells	
1.3	Recall the relative charge and relative mass of: a a proton b a neutron c an electron	
1.4	Explain why atoms contain equal numbers of protons and electrons	
1.5	Describe the nucleus of an atom as very small compared to the overall size of the atom	1d
1.6	Recall that most of the mass of an atom is concentrated in the nucleus	
1.7	Recall the meaning of the term mass number of an atom	
1.8	Describe atoms of a given element as having the same number of protons in the nucleus and that this number is unique to that element	
1.9	Describe isotopes as different atoms of the same element containing the same number of protons but different numbers of neutrons in their nuclei	
1.10	Calculate the numbers of protons, neutrons and electrons in atoms given the atomic number and mass number	3b
1.11	Explain how the existence of isotopes results in relative atomic masses of some elements not being whole numbers	1a, 1c
1.12	**Calculate the relative atomic mass of an element from the relative masses and abundances of its isotopes**	1a, 1c, 1d 3a, 3c

Use of mathematics

- Relate size and scale of atoms to objects in the physical world (1d).
- Estimate size and scale of atoms (1d).

The periodic table

Students should:		Maths skills
1.13	Describe how Dmitri Mendeleev arranged the elements, known at that time, in a periodic table by using properties of these elements and their compounds	
1.14	Describe how Dmitri Mendeleev used his table to predict the existence and properties of some elements not then discovered	
1.15	Explain that Dmitri Mendeleev thought he had arranged elements in order of increasing relative atomic mass but this was not always true because of the relative abundance of isotopes of some pairs of elements in the periodic table	
1.16	Explain the meaning of atomic number of an element in terms of position in the periodic table and number of protons in the nucleus	
1.17	Describe that in the periodic table a elements are arranged in order of increasing atomic number, in rows called periods b elements with similar properties are placed in the same vertical columns called groups	
1.18	Identify elements as metals or non-metals according to their position in the periodic table, explaining this division in terms of the atomic structures of the elements	
1.19	Predict the electronic configurations of the first 20 elements in the periodic table as diagrams and in the form, for example 2.8.1	4a 5b
1.20	Explain how the electronic configuration of an element is related to its position in the periodic table	4a

Ionic bonding

Students should:		Maths skills
1.21	Explain how ionic bonds are formed by the transfer of electrons between atoms to produce cations and anions, including the use of dot and cross diagrams	5b
1.22	Recall that an ion is an atom or group of atoms with a positive or negative charge	
1.23	Calculate the numbers of protons, neutrons and electrons in simple ions given the atomic number and mass number	3b
1.24	Explain the formation of ions in ionic compounds from their atoms, limited to compounds of elements in groups 1, 2, 6 and 7	1c 5b
1.25	Explain the use of the endings –ide and –ate in the names of compounds	

Students should:		Maths skills
1.26	Deduce the formulae of ionic compounds (including oxides, hydroxides, halides, nitrates, carbonates and sulfates) given the formulae of the constituent ions	1c
1.27	Explain the structure of an ionic compound as a lattice structure a consisting of a regular arrangement of ions b held together by strong electrostatic forces (ionic bonds) between oppositely-charged ions	5b

Use of mathematics

- Represent three dimensional shapes in two dimensions and vice versa when looking at chemical structures (5b).

Covalent bonding

Students should:		Maths skills
1.28	Explain how a covalent bond is formed when a pair of electrons is shared between two atoms	
1.29	Recall that covalent bonding results in the formation of molecules	
1.30	Recall the typical size (order of magnitude) of atoms and small molecules	1d
1.31	Explain the formation of simple molecular, covalent substances, using dot and cross diagrams, including: a hydrogen b hydrogen chloride c water d methane e oxygen f carbon dioxide	5b

Use of mathematics

- Relate size and scale of atoms to objects in the physical world (1d).
- Estimate size and scale of atoms (1d).

Types of substance

Students should:		Maths skills
1.32	Explain why elements and compounds can be classified as: a ionic b simple molecular (covalent) c giant covalent d metallic and how the structure and bonding of these types of substances results in different physical properties, including relative melting point and boiling point, relative solubility in water and ability to conduct electricity (as solids and in solution)	
1.33	Explain the properties of ionic compounds limited to: a high melting points and boiling points, in terms of forces between ions b whether or not they conduct electricity as solids, when molten and in aqueous solution	4a
1.34	Explain the properties of typical covalent, simple molecular compounds limited to: a low melting points and boiling points, in terms of forces between molecules (intermolecular forces) b poor conduction of electricity	4a
1.35	Recall that graphite and diamond are different forms of carbon and that they are examples of giant covalent substances	
1.36	Describe the structures of graphite and diamond	5b
1.37	Explain, in terms of structure and bonding, why graphite is used to make electrodes and as a lubricant, whereas diamond is used in cutting tools	5b
1.38	Explain the properties of fullerenes including C_{60} and graphene in terms of their structures and bonding	5b
1.39	Describe, using poly(ethene) as the example, that simple polymers consist of large molecules containing chains of carbon atoms	5b
1.40	Explain the properties of metals, including malleability and the ability to conduct electricity	5b
1.41	Describe the limitations of particular representations and models, to include dot and cross, ball and stick models and two- and three-dimensional representations	5b
1.42	Describe most metals as shiny solids which have high melting points, high density and are good conductors of electricity whereas most non-metals have low boiling points and are poor conductors of electricity	

Use of mathematics

- Represent three dimensional shapes in two dimensions and vice versa when looking at chemical structures, e.g. allotropes of carbon (5b).
- Translate information between diagrammatic and numerical forms (4a).

Calculations involving masses

Students should:		Maths skills
1.43	Calculate: a relative formula mass given relative atomic masses b percentage by mass of an element in a compound given relative atomic masses	1a, 1c
1.44	Calculate the formulae of simple compounds from reacting masses or percentage composition and understand that these are empirical formulae	1a, 1c 2a
1.45	Deduce: a the empirical formula of a compound from the formula of its molecule b the molecular formula of a compound from its empirical formula and its relative molecular mass	1c
1.46	Describe an experiment to determine the empirical formula of a simple compound such as magnesium oxide	1a, 1c 2a
1.47	Explain the law of conservation of mass applied to: a a closed system including a precipitation reaction in a closed flask b a non-enclosed system including a reaction in an open flask that takes in or gives out a gas	1a
1.48	Calculate masses of reactants and products from balanced equations, given the mass of one substance	1a, 1c 2a
1.49	Calculate the concentration of solutions in g dm^{-3}	1a, 1c 2a 3b, 3c
1.50	**Recall that one mole of particles of a substance is defined as:** **a the Avogadro constant number of particles (6.02×10^{23} atoms, molecules, formulae or ions) of that substance** **b a mass of 'relative particle mass' g**	1b

Students should:		Maths skills
1.51	**Calculate the number of:** **a moles of particles of a substance in a given mass of that substance and vice versa** **b particles of a substance in a given number of moles of that substance and vice versa** **c particles of a substance in a given mass of that substance and vice versa**	1a, 1b, 1c 2a 3a, 3b, 3c
1.52	**Explain why, in a reaction, the mass of product formed is controlled by the mass of the reactant which is not in excess**	1c
1.53	**Deduce the stoichiometry of a reaction from the masses of the reactants and products**	1a, 1c

Use of mathematics

- Arithmetic computation and ratio when determining empirical formulae, balancing equations (1a and 1c).
- Arithmetic computation, ratio, percentage and multistep calculations permeates quantitative chemistry (1a, 1c and 1d).
- **Calculations with numbers written in standard form when using the Avogadro constant (1b)**.
- Change the subject of a mathematical equation (3b and 3c).
- Provide answers to an appropriate number of significant figures (2a).
- **Convert units where appropriate particularly from mass to moles (1c)**.

Suggested practicals

- Investigate the size of an oil molecule.
- Investigate the properties of a metal, such as electrical conductivity.
- Investigate the different types of bonding: metallic, covalent and ionic.
- Investigate the typical properties of simple and giant covalent compounds and ionic compounds.
- Classify different types of elements and compounds by investigating their melting points and boiling points, solubility in water and electrical conductivity (as solids and in solution), including sodium chloride, magnesium sulfate, hexane, liquid paraffin, silicon(IV) oxide, copper sulfate, and sucrose (sugar).
- Determine the empirical formula of a simple compound.
- Investigate mass changes before and after reactions.
- Determine the formula of a hydrated salt such as copper sulfate by heating to drive off water of crystallisation.